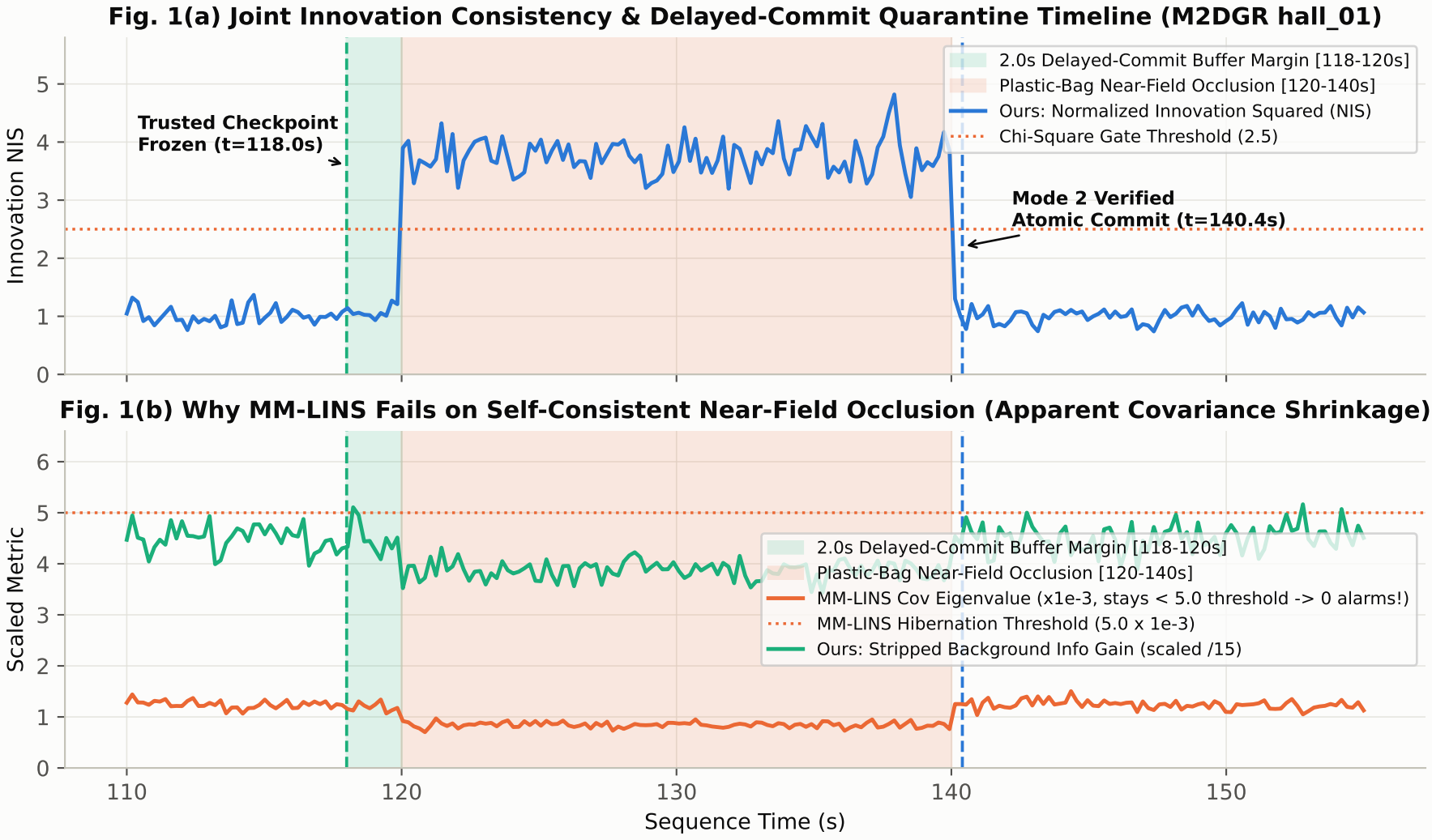
Fig. 1(a) Joint Innovation Consistency & Delayed-Commit Quarantine Timeline (M2DGR hall_01)
Innovation NIS
Trusted Checkpoint
Frozen (t=118.0s)
Mode 2 Verified
Atomic Commit (t=140.4s)
2.0s Delayed-Commit Buffer Margin [118-120s]
Plastic-Bag Near-Field Occlusion [120-140s]
Ours: Normalized Innovation Squared (NIS)
Chi-Square Gate Threshold (2.5)
Fig. 1(b) Why MM-LINS Fails on Self-Consistent Near-Field Occlusion (Apparent Covariance Shrinkage)
Scaled Metric
2.0s Delayed-Commit Buffer Margin [118-120s]
Plastic-Bag Near-Field Occlusion [120-140s]
MM-LINS Cov Eigenvalue (x1e-3, stays < 5.0 threshold -> 0 alarms!)
MM-LINS Hibernation Threshold (5.0 x 1e-3)
Ours: Stripped Background Info Gain (scaled /15)
Sequence Time (s)